

# Moving Beyond Symptom Search: How a Pre-Diagnostic Reddit Community Works Through Uncertainty

Yuting Peng

Georgia Institute of Technology  
School of Interactive Computing  
Atlanta, Georgia, USA  
ypeng390@gatech.edu

## Abstract

In this position paper, we argue that pre-diagnostic internet use should be understood as more than individual symptom searching. We examine r/doihavebreastcancer, a Reddit community organized around this period before diagnosis, through a longitudinal analysis of 15,038 threads and 113,646 comments posted between August 2022 and December 2025. Our analysis shows that the subreddit expanded rapidly while sustaining consistent levels of response, and our topic model reveals an increasingly broad set of discussions spanning bodily changes, clinical processes, risk, reassurance, and benign outcomes. Together, these patterns suggest that r/doihavebreastcancer has developed into a persistent community where questions from different points in the diagnostic process are gathered and worked through collectively. We use these findings to motivate future qualitative analysis of how participants respond to the gaps between clinical encounters. We bring these questions to the workshop to consider what kinds of technological support might strengthen the relationships and forms of judgment already present in such communities.

## Keywords

online health communities, women’s health, Reddit, topic modeling

## 1 Introduction

Experiences of possible breast cancer vary widely, spanning the discovery of bodily changes, diagnostic imaging and biopsy and the interpretation of clinical findings. On r/doihavebreastcancer, people post condensed ultrasound impressions, describe a newly noticed lump or skin change, compare BI-RADS categories, and ask how worried they should be about what happens next. This is not primarily a community for coping after a diagnosis. It is organized around an open question, namely whether a bodily change, scan, or report may indicate breast cancer. An AutoModerator reply routinely reminds posters that community members are not doctors and cannot diagnose them. Despite this, folks continue to submit questions to this subreddit, moving from what a symptom looks or feels like to what an imaging report means, how painful a biopsy may be, or how long results usually take.

Research on online health communities has shown how people exchange emotional support, experiential knowledge, and practical advice [3, 8, 10, 13, 15]. Most of this work examines communities organized around an established condition or a recognizable identity such as patient, survivor, caregiver, or long-term condition manager [8]. There is an opportunity to understand pre-diagnostic uses of online communities. A person may be deciding whether a sensation is worth reporting, waiting for imaging, or trying to

understand a clinical document without knowing whether they will ever become a cancer patient. Studies of breast irregularities and other pre-diagnostic journeys show that bodily changes become meaningful through interpretation, social comparison, and encounters with healthcare systems [2, 5, 9]. However, research on web-based symptom appraisal has often treated internet use as an individual search activity that may guide help-seeking or intensify concern [11, 14].

Our case suggests that this individual-search framing is incomplete. People do not only query a search engine and leave. They also post in a continuing community where new questions accumulate, receive replies, and remain available to later visitors. The community therefore offers a useful empirical case for asking what concerns collect around breast health before diagnosis and how those concerns change as the community grows. This position paper reports two completed quantitative analyses, a longitudinal account of community growth and a topic model of the full corpus. We do not use these analyses to claim that we already know how support is produced inside threads. Instead, we use them to establish how much participation the community attracted, what kinds of questions appeared, and how their balance shifted over time.

This distinction also helps situate our contribution to the workshop. Before asking how AI might reorganize women’s health work, a clearer account of where uncertainty already gathers and what people repeatedly seek from one another can help us better understand future opportunities for design. The following results may provide that map while leaving the design questions open for subsequent qualitative study and workshop discussion.

## 2 From Occasional Questions to a Sustained Community

Our dataset spans the subreddit’s creation in August 2022 through December 31, 2025. It contains 15,038 submissions and 113,646 comments from a community-maintained Academic Torrents archive [4, 12]. We reconstructed each thread by linking its submission and full comment tree, then removed 18,987 AutoModerator comments, one repeated bot-template comment, one policy/admin comment, 1,085 deleted comments, and 321 removed comments. The resulting corpus contains 93,251 retained comments and 15,037 non-empty thread documents. Because the posts concern bodily fears and medical uncertainty, we report only aggregate results and follow established ethical guidance for Reddit and internet research [6, 7].

We first calculated quarterly numbers of new threads, retained comments, and comments per thread. The growth pattern is not a smooth increase but shows three recognizable periods. After the partial launch quarter, the community received 199 new threads

in 2022 Q4. During 2023, quarterly volume increased from 333 to 628 threads, indicating that the subreddit was becoming more than a small set of occasional posts. A larger expansion occurred in 2024, when volume passed 1,000 threads in Q1, reached 1,717 in Q3, and remained at 1,689 in Q4. The community peaked at 2,076 new threads and 13,834 retained comments in 2025 Q2. Although activity then declined to 1,570 threads in 2025 Q4, that final quarter still contained nearly eight times as many threads as the first complete quarter.

Higher volume did not coincide with an obvious collapse in reply levels. Across complete quarters, the mean number of retained comments per thread ranged from 4.6 to 6.9. The median was three in 2022 Q4, four from 2023 Q1 through 2025 Q2, and three in 2025 Q3 and Q4. In other words, as quarterly posting increased more than tenfold, a typical thread continued to attract several non-automated comments. This shows that growing demand was accompanied by continuing response activity.

The subreddit became a sustained destination where many people brought uncertainty into a shared, publicly visible space, and where the volume of replies kept pace with much of the growth. The next question is about what kinds of uncertainty filled that expanding space. The quarterly growth patterns and topic trends described in the next section can also be explored interactively through a visualized website.<sup>1</sup>

### 3 From Symptom Appraisal to a Broader Diagnostic Landscape

To map recurring concerns, we concatenated each submission with its retained comments and used latent Dirichlet allocation [1]. We trained models with 9, 10, and 11 topics and evaluated each using coherence scores ( $c_v$ ) and qualitative inspection of the top words and high-probability documents. The ten-topic model offered the best balance of topic distinctiveness and interpretability, so we adopted it for subsequent analysis. Topic labels were assigned through examination of top words and representative documents to aid interpretation. The findings reported here remain corpus-level counts and trends.

The ten labels (Table 1) are best understood as recurring centers of discussion. A question about imaging may also involve waiting or reassurance, and a benign outcome may be narrated through the symptom or test that preceded it.

The first centers on bodily changes. Discovering lumps, noticing skin or nipple changes, and feeling armpit pain together account for 40.1% of all threads. These topics capture how uncertainty often begins with interpreting a newly noticed sensation or visible change, sometimes before any clinical contact.

A second group concerns the formal diagnostic process. Imaging, clinical reports, waiting, and biopsy together account for 31.5% of all threads. The vocabulary of these topics includes mammograms, ultrasound, MRI, dense tissue, BI-RADS categories, masses, calcifications, appointments, procedures, pain, and result timelines. This pattern shows that clinical contact often changes the form of uncertainty rather than resolving it. A bodily concern may become a report full of unfamiliar terminology. An imaging appointment

may produce a recommendation for additional views. A biopsy may replace uncertainty about a symptom with uncertainty about the procedure and its results. The topic model, therefore, makes visible a chain of questions that extends across bodies, tests, documents, and waiting periods.

A third group concerns possible meanings and outcomes. Benign findings are the third-largest individual topic at 11.8%, while reassurance and risk account for another 16.6%. This is analytically important because *r/doihavebreastcancer* is not a collection of confirmed cancer cases. Many trajectories end with cysts, fibroadenomas, benign imaging, routine surveillance, or no reported conclusion. The large benign topic shows that non-cancer outcomes are a substantial part of what later readers encounter when trying to interpret their own situation. At the same time, the separate reassurance and risk topics indicate two different orientations toward uncertainty, one seeking reasons not to panic and the other seeking reasons a case might deserve closer attention.

The topic proportions changed as the community matured. In 2022 Q4, lumps and skin or nipple changes made up 42.7% of threads, whereas reading clinical reports represented only 1.0%. By 2025 Q4, the two symptom topics had fallen to 25.4%, while clinical reports, biopsy, and risk together had risen from 8.0% to 22.8%. From 2024 onward, no single topic accounted for more than 17.1% of any quarter. Discussion became more evenly distributed across different points in the diagnostic process as the community matured. People also arrived after entering care, carrying new uncertainties produced by medical terminology, procedural recommendations, family history, and waiting.

### 4 Future Directions

This position paper has mapped how *r/doihavebreastcancer* grew and what concerns repeatedly appeared, but topic modeling cannot show what commenters actually did with those concerns. Our next step is a qualitative analysis of sampled full threads. We will examine what posters ask for, how replies use personal experience, how members refer one another toward or away from professional care, how uncertainty is expressed and calibrated, and when a conversation reassures, escalates concern, or remains unresolved. Analyzing the submission together with its full comment tree will also allow us to distinguish a topic from the response process around it. The same imaging report, for example, could prompt explanation, comparison, emotional support, disagreement, or no substantive answer.

This next phase is also necessary before making strong design claims. The quantitative results identify recurring pressure points, including terminology that is hard to parse, timelines that are hard to track, and procedural steps that produce their own anxiety. AI might help a person prepare questions for an appointment or make sense of clinical terminology, but it could also produce false certainty or blur the line between peer experience and automated output. Any design proposal should therefore begin by understanding the existing work of posters and commenters and by preserving clear routes to professional care and correction.

Several questions from this study would benefit from discussion at the workshop.

<sup>1</sup>An interactive visualization of quarterly growth and topic trends is available at <https://kitepretty.github.io/pre-diagnostic-liminal-care-dashboard/>.

**Table 1: Ten LDA topics with representative top words, thread counts, and corpus share ( $N = 15,037$ ).**

| Topic                        | Representative words                                               | Threads | Share |
|------------------------------|--------------------------------------------------------------------|---------|-------|
| Discovering lumps            | lump, ultrasound, feel, lumps, hard, cyst, tissue                  | 2,244   | 14.9% |
| Noticing skin/nipple changes | nipple, skin, red, ibc, rash, discharge, noticed                   | 2,085   | 13.9% |
| Receiving benign findings    | benign, birads, fibroadenoma, mass, hope, news, radiologist        | 1,778   | 11.8% |
| Feeling armpit pain          | pain, lymph, armpit, nodes, swollen, symptoms, months              | 1,695   | 11.3% |
| Seeking reassurance          | anxiety, sorry, hope, waiting, need, feel, think                   | 1,690   | 11.2% |
| Navigating imaging           | mammogram, ultrasound, mri, dense, imaging, diagnostic, follow     | 1,397   | 9.3%  |
| Reading clinical reports     | ultrasound, cm, mass, birads, findings, suspicious, calcifications | 1,338   | 8.9%  |
| Waiting for results          | results, wait, waiting, appointment, called, days, week            | 1,102   | 7.3%  |
| Undergoing biopsy            | biopsy, procedure, needle, day, ice, experience                    | 900     | 6.0%  |
| Managing risk                | risk, surgery, surgeon, lumpectomy, genetic, family, history       | 808     | 5.4%  |

- How do different forms of pre-diagnostic uncertainty (e.g., unfamiliar bodily changes, clinical terminology, procedural waiting) shape what community members seek from one another, and how do peer responses relate to formal care pathways?
- Beyond individual posters and commenters, what roles do other actors (e.g. platform features, automated messages) play in shaping how pre-diagnostic health communities handle clinical questions they cannot answer?
- Where might AI support people in organizing or accessing health information within communities like this one without turning uncertain peer knowledge into diagnosis or replacing the judgment that members already provide?

Participation in the Women's Health as Cooperative Work workshop would allow us to share this longitudinal case with researchers studying women's health across clinics, communities, and digital platforms. We hope to compare this pre-diagnostic setting with other health contexts, receive feedback on the qualitative phase, and develop an account of AI that remains anchored in the questions people already bring, the limits of peer discussion, and the responsibilities of formal care.

## References

- [1] David M. Blei, Andrew Y. Ng, and Michael I. Jordan. 2003. Latent dirichlet allocation. *J. Mach. Learn. Res.* 3, null (March 2003), 993–1022.
- [2] Susanne Brandner, Jacqueline Müller-Nordhorn, Wiebke Stritter, Christina Fotopoulou, Jalid Sehoul, and Christine Holmberg. 2014. Symptomization and Triggering Processes: Ovarian Cancer Patients' Narratives on Pre-Diagnostic Sensation Experiences and the Initiation of Healthcare Seeking. *Social Science & Medicine* 119 (Oct. 2014), 123–130. doi:10.1016/j.socscimed.2014.08.022
- [3] Shaan Chopra, Rachael Zehrung, Tamil Arasu Shanmugam, and Eun Kyoung Choe. 2021. Living with Uncertainty and Stigma: Self-Experimentation and Support-Seeking around Polycystic Ovary Syndrome. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, 1–18. doi:10.1145/3411764.3445706
- [4] Joseph Paul Cohen and Henry Z. Lo. 2014. Academic Torrents: A Community-Maintained Distributed Repository. In *Proceedings of the 2014 Annual Conference on Extreme Science and Engineering Discovery Environment (XSEDE '14)*. Association for Computing Machinery, New York, NY, USA, 1–2. doi:10.1145/2616498.2616528
- [5] Catherine Cook, Margaret Brunton, Tepora Pukepuke, and Ai Ling Tan. 2018. Exploring Communication during the Journey from Noticing Bodily Changes to a Diagnosis of Endometrial Cancer. *Journal of Clinical Nursing* 27, 5–6 (March 2018), 1262–1275. doi:10.1111/jocn.14173
- [6] Casey Fiesler, Michael Zimmer, Nicholas Proferes, Sarah Gilbert, and Naiyan Jones. 2024. Remember the Human: A Systematic Review of Ethical Considerations in Reddit Research. *Proc. ACM Hum.-Comput. Interact.* 8, GROUP (Feb. 2024), 5:1–5:33. doi:10.1145/3633070
- [7] Aline Shakti Franzke, Anja Bechmann, Michael Zimmer, Charles M. Ess, and Association of Internet Researchers. 2020. *Internet Research: Ethical Guidelines 3.0*. Technical Report. Association of Internet Researchers. <https://aoir.org/reports/ethics3.pdf>
- [8] Doğa Gatos, Aslı Günay, Güncel Kırılancık, Kemal Kusu, and Asim Evren Yantac. 2021. How HCI Bridges Health and Design in Online Health Communities: A Systematic Review. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference (DIS '21)*. Association for Computing Machinery, New York, NY, USA, 970–983. doi:10.1145/3461778.3462100
- [9] Leeat Granek and Karen Fergus. 2012. Resistance, Agency, and Liminality in Women's Accounts of Symptom Appraisal and Help-Seeking upon Discovery of a Breast Irregularity. *Social Science & Medicine* 75, 10 (Nov. 2012), 1753–1761. doi:10.1016/j.socscimed.2012.07.016
- [10] Jina Huh and Mark S. Ackerman. 2012. Collaborative Help in Chronic Disease Management: Supporting Individualized Problems. In *Proceedings of the ACM 2012 Conference on Computer Supported Cooperative Work (CSCW '12)*. Association for Computing Machinery, New York, NY, USA, 853–862. doi:10.1145/2145204.2145331
- [11] Julia Mueller, Caroline Jay, Simon Harper, Alan Davies, Julio Vega, and Chris Todd. 2017. Web Use for Symptom Appraisal of Physical Health Conditions: A Systematic Review. *Journal of Medical Internet Research* 19, 6 (June 2017), e202. doi:10.2196/jmir.6755
- [12] stuck\_in\_the\_matrix, Watchful1, and RaiderBDev. 2026. Reddit Comments/Submissions 2005-06 to 2025-12. Academic Torrents. Data set. <https://academictorrents.com/details/3d426c47c767d40f82c7ef0f47c3acacdd2bf44>
- [13] Tatiana A. Vlahovic, Yi-Chia Wang, Robert E. Kraut, and John M. Levine. 2014. Support Matching and Satisfaction in an Online Breast Cancer Support Community. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1625–1634. doi:10.1145/2556288.2557108
- [14] Ryan W. White and Eric Horvitz. 2009. Cyberchondria: Studies of the Escalation of Medical Concerns in Web Search. *ACM Trans. Inf. Syst.* 27, 4 (Nov. 2009), 23:1–23:37. doi:10.1145/1629096.1629101
- [15] Alyson L. Young and Andrew D. Miller. 2019. "This Girl Is on Fire": Sensemaking in an Online Health Community for Vulvodynia. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300359